Module 12: Darwin and Evolution — Questions

Practice Questions

1. From his observations in the Galápagos Islands, what did Darwin conclude about whether species can change over time and how they relate to their environments?
2. In evolutionary biology, what does "fitness" mean? Why is it *not* the same as physical strength or being "healthy" in everyday language?
3. In biology, natural selection acts on variation in traits within a population. Does it change individuals during their lifetimes, or change allele frequencies over generations? Explain in one or two sentences.
4. The forelimbs of a human, a cat, a whale, and a bat share the same underlying bone pattern but serve different functions. What kind of structures are these, and what do they suggest about common ancestry?
5. How are analogous structures different from homologous structures? Give a brief example of each type (you may use different examples than the book).
6. Is natural selection a random process? Explain what is and is not random when natural selection occurs.
7. Why is the phrase "survival of the fittest" often misleading to the public? Restate what "fitness" means in evolutionary terms.
8. A beetle population includes green and brown individuals in a lush green forest; birds eat more brown beetles because they are easier to see. Later, drought browns the vegetation. Describe step-by-step how natural selection could shift the population's color over generations.
9. How do comparisons of DNA and proteins support the idea of common ancestry and evolution?
10. What is meant by "descent with modification" in Darwin's terms?
11. How does artificial selection (for example, breeding plants or livestock) illustrate principles that also apply to natural selection?
12. What is a vestigial structure, and why do biologists treat vestigial traits as evidence of evolutionary history?

13. What is convergent evolution, and how does it produce similarities that are *not* due to recent common ancestry?
14. In what ways does the fossil record support the idea that life on Earth has changed over long periods of time?
15. In modern genetics, why do acquired traits during an organism's life (for example, muscle built from exercise) not explain how populations evolve across generations?